

PHS-SEC-1-1 : Introduction to Computer Programming and Graph Plotting

(Identical content with PHS-CC-SEC-1-1)

Practical [4 Credits, 60 lab periods]

1. Introduction to Graph Plotting (2D only, using Gnuplot) •

(a) Plotting 2D graphs: both functions and data files. Changing plot range and plot styles: the options: with points (w p), with dots (w d), with lines (w l), with linespoints (w lp), linestyle (lt), linewidth (lw). Using the set command for samples, xrange, yrange, xlabel, ylabel, title etc. The *using* option.

(b) User defined functions [Including the use of ternary operator (? :) for piece-wise defined functions.]

(c) Fitting data files using Gnuplot.

(d) Polar and parametric plots

(e) Conditional plotting of data from file using \$, &&, || operators. (Graphs to be saved without using GUI)

2. Introduction to programming in Python (Version 3) •

(a) Introduction

Using the python interpreter as a calculator

Variable and data types (int, float, complex, list, tuple, set, string, the type () function)

Basic mathematical operations

Compound statements in python

– Conditionals (if, elif, else)

– Loops (for, while)

– User defined functions def: (return statement, default values for arguments, keyword arguments), lambda function.

Importing modules with math and cmath as examples

Using help and dir command to use the inbuilt manual

Basic idea of namespaces – local and global

Python scripts, I/O operations (including opening and writing to files)

(b) The python data types

List: defining lists, reading and changing elements from lists, slicing (with discussion on the difference between $ll=mm$ and $ll=mm[:]$, concatenation, list comprehension

– built in functions involving lists: range(), len(), sum(), min(), max() – list methods: append(), extend(), count(), index(), sort(), insert(), pop(), remove(), reverse()

Tuples: Contrast and compare with lists, packing/unpacking using tuples (including $a, b = b, a$ to swap variables)

Sets : set methods: update(), pop(), remove(), Set Theoretic operations: union, intersection, difference and symmetric difference of two sets

Strings: defining strings, the use of single, double or triple quotes as string delimiters, len(), indexing, slicing, string concatenation, some string methods: strip(), split(), join(), find(), count(), replace(), string formatting in python (using the

3. Problems and Applications •

Finding factors of an integer

Determining whether an integer is prime or not

Finding out prime number greater than or lesser than a given value
 Finding out all prime numbers within a given range
 Root finding for a single variable (basic theory and algorithm) using Newton-Raphson and Bisection method
 Sorting of lists (algorithm, flowchart and code) using Bubble or Selection sort
 Sum of series correct up to given decimal places (Sine, Cosine, Exponential etc.)
 Simulation of motion of a particle in 1D under a given force $F(x, t, v)$ with given initial condition and plotting $(x, t), (x, v), (t, v)$. (Output to be saved in data files and Gnuplot to be used to plot graphs), using Euler's method only
 Matrix Addition, Multiplication and Transpose using List Comprehension

Recommended reading

1. Gnuplot in Action understanding data and Graphs, Phillipp K. Janert
2. Scientific Computing in Python. Abhijit Kar Gupta, Techno World
3. Computational Physics, Mark Newman, Amazon Digital.
4. Physics in Laboratory including Python Programming (Semester I), Mandal, Chowdhury, Das, Das, Santra Publication
5. Introduction to Numerical Analysis, S.S. Sastry, 5th Edn. , 2012, PHI Learning Pvt. Ltd
6. Numerical Mathematical Analysis, J. B. Scarborough, OXFORD and IBH Co. Pvt. Ltd.
7. Elementary Numerical Analysis, K.E. Atkinson, 3rd Edn., 2007, Wiley India Edition
8. Gnuplot 5, Lee Phillips, Alogus Publishing, edition 2012.